YEAR 7 · UNIT 1 · HOMEWORK 3

Factors, Tests and Tissues

Mathematics 1.4 — Highest Common Factors · 1.5 — Tests for Divisibility
Science 1.4 — Cells, Tissues and Organs

Name: ______________________ **Class:** ______________________

Date given: ____________________ **Date due:** ____________________

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the division, not just the answer**. Almost every question in Section A and Section B is really a division question.
- In science, answer in **full English sentences**. "Organ" is not a sentence; "The heart is an organ" is. If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A · Mathematics 1.4 — Highest Common Factors

Remember from the lesson

A **factor** of a number divides into it **exactly**, with nothing left over. You find factors by hunting for **pairs**: $24 = 1 \times 24$, 2×12, 3×8, 4×6.

Common means **shared**. A **common factor** is in both lists, and the **highest common factor (HCF)** is the biggest of them — the same idea you already know as **UCLN**.

The HCF is never "none". 1 is a factor of every number, so every pair has at least one common factor.

A1 Hunt the pairs

(a) Write **every** pair of whole numbers that multiplies to **36**, then list all the factors of 36.

Pairs: ____

Factors of 36: ____

(b) Now do the same for **54**.

Pairs: ____

Factors of 54: ____

(c) Which numbers are in **both** lists? ____________________________________

(d) So the **HCF** of 36 and 54 is ____________

A2 Divide to decide

The only way to be sure

There is one test that always works: **do the division**. If there is **nothing left over**, it is a factor. If there is a **remainder**, it is not.

Do each division. Write the answer, and the remainder if there is one. Then tick the box only if the first number is a **factor**.

Division	Answer	Factor?	Division	Answer	Factor?
$96 \div 6$		☐	$96 \div 7$		☐
$84 \div 4$		☐	$84 \div 5$		☐
$105 \div 7$		☐	$132 \div 8$		☐
$156 \div 12$		☐	$210 \div 6$		☐

A3 Say it, then write it

Word help — the English tells you which number to start with

divide 96 by 8	you **start** at 96: $96 \div 8$.
share 72 equally between 9	divide: $72 \div 9$.
6 divides into 84	$84 \div 6$, not $6 \div 84$.
exactly	with **nothing left over**. No remainder.
the highest common factor of	the **biggest** number that divides into **both**.
in its simplest form	written with the **smallest** numbers you can.

Turn each sentence into a calculation, then work it out. **Write the calculation first.**

1. Divide 96 by 8.
   Calculation: ______________________ Answer: __________

2. Share 72 equally between 9.
   Calculation: ______________________ Answer: __________

3. Find the highest common factor of 20 and 30.
   Calculation: ______________________ Answer: __________

4. Does 6 divide into 81 exactly? Show the division.
   Calculation: ______________________ Answer: __________

A4 What the HCF is actually for

Simplest form in one step

Find the HCF of the top and the bottom, then **divide both of them by it**. That is the whole method — and it is two more divisions.

Write each fraction in its simplest form. Fill in every box.

Fraction	HCF	Top ÷ HCF	Bottom ÷ HCF	Simplest form
$\frac{18}{24}$				
$\frac{30}{45}$				
$\frac{48}{60}$				

A5 Two answers that surprise people

Work these out before you read on

Both of these pairs catch nearly everybody. Do the factor lists if you need to.

(a) The HCF of **9 and 10** is ____________

(b) The HCF of **7 and 42** is ____________

(c) A student writes that 9 and 10 have **no** common factor. In one full sentence, explain why that can never be true for any pair of numbers.

A6 Word problems

Read each one **all the way to the end** before you start writing.

1. **The bracelets.** Mr Bowen has **42 red beads** and **56 blue beads**. He makes identical bracelets from them, using every bead, with **nothing left over**, and he wants **as many bracelets as possible**. How many bracelets can he make, and how many beads of each colour go on one bracelet?

2. **The test.** There are **60 questions** on a test. Mr Bowen gets **45** of them right. Write the

fraction he got right, in its **simplest form**.

Section B · Mathematics 1.5 — Tests for Divisibility

The nine tests you wrote down

by 2 the last digit is 0, 2, 4, 6 or 8
by 3 the digits add up to a multiple of 3
by 4 the last **two** digits make a multiple of 4
by 5 the last digit is 0 or 5
by 6 it passes the test for 2 **and** the test for 3
by 8 the last **three** digits make a multiple of 8
by 9 the digits add up to a multiple of 9
by 10 the last digit is 0
by 11 split the digits into two groups, taking every other one, and add each group. The **difference** is 0 or a multiple of 11

There is no easy test for 7. For 7 you divide and look at the remainder.

B1 The factor hunt

Use the tests. Put a **tick** in the box if the number on the left divides by the number at the top, and a **cross** if it does not. Do **not** use a calculator.

	2	3	4	5	6	8	9	10	11
1260									
3465									
2088									
4510									
5832									

Now pick **one** of your ticks and say which test proved it.

I know that _________ divides by ______ because ____________________________

B2 Now prove it

A test says yes or no. Division says how many.

A divisibility test is a **shortcut**: it tells you there is nothing left over, but it never tells you the **answer**. For that you still have to divide.

Each of these came out as a tick in B1. Carry out the division and write the answer.

(a) $1260 \div 4 =$ __________ (d) $2088 \div 8 =$ __________

(b) $1260 \div 9 =$ __________ (e) $4510 \div 5 =$ __________

(c) $3465 \div 11 =$ __________ (f) $5832 \div 6 =$ __________

(g) Two of those six have the **same** answer. Which two? ______________________________

B3 The missing digit

In each one a digit has been rubbed out. Use a test to work out what it must be. Show the digit sum or the groups you used.

1. 5 2 □ is divisible by **3**. There is more than one answer — find them all.
Working: ________________________________ Digit(s): ______________

2. 7 4 1 □ is divisible by **9**.
Working: ________________________________ Digit: ______________

3. □ 6 3 0 is divisible by **11**.
Working: ________________________________ Digit: ______________

B4 Word problems

1. The stickers. Mr Bowen has **4176 stickers** to share equally between the **8** classes in the school. Use a test to decide whether he can do it with none left over — then work out how many each class gets.

2. The socks. Mr Bowen has **2464 socks**. He puts them into **pairs**, with none left over, and then puts the pairs into drawers of **11 pairs** each, again with none left over. Show that he can do both, and work out how many drawers he needs.

B5 Now you write the question

Your turn to be the teacher

So far the questions have given you the English and you have written the maths. Now do it **backwards**.

Your word problem must:

- be about something **real**, written in **full English sentences**, and end with a question;
- share things out **equally with nothing left over**;
- have the answer that is already given to you.

(a) Write a word problem for $4368 \div 8 = 546$

(b) How could you tell that 4368 divides by 8 **without** doing the division? Answer in one full sentence.

Section C · Science 1.4 — Cells, Tissues and Organs

Remember from the lesson

The word that does all the work in this section is **similar** against **different**. **Similar** cells make a **tissue**. **Different** tissues make an **organ**.

C1 Match the word to its meaning

Write the correct letter in the box next to each word.

Word	Letter	Meaning
1. Tissue	☐	**A** A living thing, which may contain many organ systems.
2. Organ	☐	**B** A group of similar cells, all doing one particular job.
3. Organ system	☐	**C** A structure made of several different tissues working together.
4. Organism	☐	**D** A set of organs that all work together to do the same job.
5. Epithelium	☐	**E** The tall tissue near the top of a leaf, packed with chloroplasts.
6. Palisade layer	☐	**F** A tissue that covers a surface.

C2 Which level is it?

Write **cell**, **tissue**, **organ**, **organ system** or **organism** for each one.

What it is	Which level?	What it is	Which level?
a red blood cell		the digestive system	
a leaf		onion epidermis	
the palisade layer		Mr Bowen	
the heart		a root hair cell	

Two of those eight catch almost everybody: **a leaf** and **the palisade layer**. In one full sentence each, say how you decided.

A leaf is ____

The palisade layer is ____

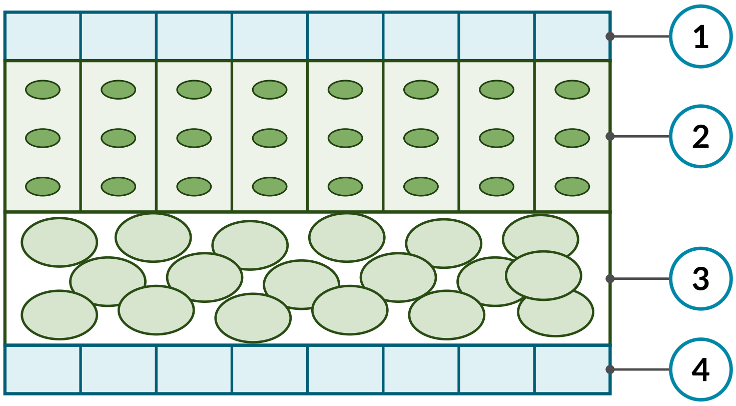
C3 Climb the ladder

One example, climbed all five rungs. The first is done for you.

Level	The example
Cell	a ciliated cell
Tissue	
Organ	
Organ system	
Organism	

C4 Label the leaf

A leaf is one **organ** made of **four different tissues**, cut open here. Name each tissue and say what it does.



No.	Name of the tissue	Its job
1		
2		
3		
4		

C5 Answer in full sentences

Word help — one word, two completely different meanings

a tissue (everyday) a paper handkerchief. You can count them: *a tissue*, *two tissues*.

tissue (science) a group of similar cells. You do **not** count it: *muscle tissue*, not *a muscle tissue*.

1. Write **two sentences of your own**. In the first, use *tissue* with its everyday meaning. In the second, use *tissue* with its scientific meaning.

2. A jellyfish has no heart, no lungs and no brain. Which level of the ladder can a living thing manage without? Explain your answer.